

6(a)(i)	<i>any four from:</i> 1 random / spontaneous ; 2 change in, base / nucleotide, sequence of DNA ; 3 ref. to base, substitution / deletion / addition ; 4 ref. to frame shift / AW ; 5 AVP ; e.g. mutagens / UV light / ionising radiation	4
6(a)(ii)	allele only expressed, when a dominant allele not present / in a homozygote or allele not expressed in a heterozygote ;	1
6(a)(iii)	symbols explained ; e.g. A = <u>allele</u> for, melanin production / normal pigment a = <u>allele</u> for, no melanin production / albinism parental genotypes ; e.g. Aa and Aa offspring genotypes identifying child with albinism as aa ; e.g. (AA Aa Aa) aa	3

Question	Answer	Marks
6(b)	<i>any three from:</i> 1 the greater the concentration of extract, the lower the activity of tyrosinase ; A inversely proportional 2 extract acts as an inhibitor / enzyme inhibited ; 3 binds to, active site / allosteric site, (of tyrosinase) ; 4 ref. to alters pH ; 5 extract denatures tyrosinase ;	3